Supplementary Material

Acoustic drivers of infant attention in infant-directed speech [Stage 1 Registered Report]

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Overview

This document is the supplementary material for the registered report *Acoustic drivers of infant attention in infant-directed speech*. It covers two topics: stimulus preparation and design, including the derivation of population norms for acoustic manipulation and the counterbalancing scheme; and a simulation-based power analysis. The study uses a 2×2×2 fully crossed within-subjects design to examine the effects of three independently manipulated acoustic properties of infant-directed speech (IDS) on total fixation time (ms) in pre-linguistic infants aged 3–9 months: mean fundamental frequency (f_0 mean) and pitch variability (f_0 SD) are the two confirmatory parameters (H1 and H2), and formant frequencies (proportional scaling of all formant frequencies, targeting apparent vocal tract length) are treated as an exploratory parameter.

The primary statistical model is a linear mixed model (LMM):

$$\texttt{fixation\_time} \sim f_0\ \text{SD} \times f_0\ \text{mean} \times \texttt{formant} + \texttt{block} + \texttt{avatar} + (1 + \texttt{avatar} \mid\mid \text{ID}) + (1 \mid \text{original})$$

The simulation uses a simplified version of this model ($\texttt{fixation\_time} \sim f_0$ SD $\times$ f_0 mean $\times$ $\texttt{formant}$ + (1 | ID)) which omits the additional fixed and random effects. This simplification is conservative for the reasons described at the end of this section.

Three complementary analyses are presented here:

1. **Frequentist detection power** (Monte Carlo, H_1 population): the probability of detecting each main effect via a significant *p*-value (α = .05) in the LMM ANOVA, as a function of sample size. Estimated from a superpopulation incorporating all expected effects.
2. **Frequentist equivalence power** (Monte Carlo, H_0 population): the probability of concluding practical equivalence with zero (i.e., the probability that a 90% CI from the marginal `emmeans` contrast falls entirely within pre-specified equivalence bounds of ±150 ms) as a function of sample size. Critically, this is estimated from a **separate** superpopulation in which all acoustic effects are set to zero, because using the H_1 data would yield near-zero equivalence rates by design (the CI stays outside the bounds precisely because real effects exist).
3. **Bayesian sensitivity analysis** (brms + ROPE): a number of Bayesian LMMs fitted in `brms` at the target sample size (n = 200) under each population. The proportion of the posterior for each main effect falling within the Region of Practical Equivalence (ROPE: ±150 ms) is computed using `bayestestR::equivalence_test()`, serving as a secondary robustness check.

The simulation assumes one observation per condition per participant and a single stimulus source. The actual design includes two blocks, yielding two observations per condition per participant, which halves the variance of each `emmeans` contrast, reducing the SE by a factor of approximately $\sqrt{2}$ relative to the simulated design. The simulation additionally omits random effects for original voice recordings (1 | original) and the by-participant avatar slope (1+`avatar` || ID), which introduce additional variance not captured here. The net power estimates are therefore conservative: the doubling of within-participant observations substantially outweighs the additional variance from the expanded random-effects structure, and the reported power curves should be treated as lower bounds on the power of the actual design.

Reproducibility. All code, seeds, and parameters are documented here. The complete source file is available at: https://doi.org/10.17605/OSF.IO/D7V3E.

1 Stimuli and Design

1.1 Stimulus Counterbalancing

The table below shows the assignment of acoustic conditions to original recordings (A–D) and avatars across the four counterbalancing lists (G1–G4) and two blocks. Within each block, every original appears exactly twice, paired with different acoustic conditions. Across lists, each acoustic condition is paired equally often with each original. Block 2 mirrors Block 1 with avatars swapped, fully counterbalancing avatar at the within-participant level.

Table S1. Assignment of acoustic conditions to original recordings and avatars across counterbalancing lists and blocks.

f_0 SD	f_0 mean	Formant freq.	G1	G2	G3	G4
			Block 1			
Low	Low	Low	A (Duck)	B (Bear)	C (Duck)	D (Bear)
Low	Low	High	B (Bear)	C (Duck)	D (Duck)	A (Bear)
Low	High	Low	C (Duck)	D (Bear)	A (Bear)	B (Duck)
Low	High	High	D (Bear)	A (Duck)	B (Bear)	C (Duck)
High	Low	Low	A (Bear)	C (Duck)	D (Bear)	B (Duck)
High	Low	High	B (Duck)	D (Bear)	A (Bear)	C (Duck)
High	High	Low	C (Bear)	A (Duck)	B (Duck)	D (Bear)
High	High	High	D (Duck)	B (Bear)	C (Duck)	A (Bear)
			Block 2			
Low	Low	Low	A (Bear)	B (Duck)	C (Bear)	D (Duck)
Low	Low	High	B (Duck)	C (Bear)	D (Bear)	A (Duck)
Low	High	Low	C (Bear)	D (Duck)	A (Duck)	B (Bear)
Low	High	High	D (Duck)	A (Bear)	B (Duck)	C (Bear)
High	Low	Low	A (Duck)	C (Bear)	D (Duck)	B (Bear)
High	Low	High	B (Bear)	D (Duck)	A (Duck)	C (Bear)
High	High	Low	C (Duck)	A (Bear)	B (Bear)	D (Duck)
High	High	High	D (Bear)	B (Duck)	C (Bear)	A (Duck)

1.2 Stimulus Generation

Eight manipulated versions of each original IDS recording will be generated using a custom Praat script (Boersma & Weenink, 2026) (available at https://doi.org/10.17605/OSF.IO/H5SNB). Each version corresponds to one cell of the fully crossed 2×2×2 design, in which mean fundamental frequency (f_0 mean), pitch variability (f_0 SD), and formant frequencies are each set to a High or Low level.

The two levels for each parameter are defined as ±1 between-speaker standard deviation (SD) from the IDS population distribution, computed from Hilton et al.'s (2022) cross-cultural corpus of female IDS ($n = 21$ societies; audio: https://doi.org/10.5281/zenodo.5525161). Using a single large cross-cultural dataset as the sole source of population norms follows the approach of Aung et al. (2024), who used ERB-unit steps calibrated to population norms for cross-cultural voice manipulation. The manipulation targets for each speaker are:

$$\text{target}_{\text{high}} = \text{baseline}_{\text{speaker}} + \text{SD}_{\text{population}}, \quad \text{target}_{\text{low}} = \text{baseline}_{\text{speaker}} - \text{SD}_{\text{population}}$$

The resulting offsets are:

- f_0 **mean:** ±51 Hz (1 × 50.8 Hz; between-speaker SD from Hilton et al. (2022), population mean f_0 = 261.8 Hz). Cross-checked against Cox et al. (2023) (meta-analytic OSF data, n = 202 female speakers; SD = 40.1 Hz), whose smaller value is consistent with the lower cross-cultural diversity of that dataset.
- f_0 **SD:** ±28 Hz (1 × 28.4 Hz; between-speaker SD from Hilton et al. (2022), population mean f_0 SD = 66.6 Hz). Cross-checked against Broesch and Bryant (2015) (Table 1; Western SD = 24.0 Hz, non-Western SD = 22.4 Hz), both consistent with the Hilton et al. value.
- **Formant frequencies:** ±4.6% proportional scaling (1× CV = 0.046). F1–F4 were extracted from the Hilton et al. (2022) corpus audio using a custom Praat script; the between-speaker SD of the geometric mean of F1–F4 was 87.4 Hz on a population mean of 1920.9 Hz (CV = 0.046). This replaces a provisional pilot estimate and provides a principled population anchor for the formant manipulation; however, formant frequencies remain exploratory for the reasons detailed in the main manuscript (sparse literature on infant attentional responses to this parameter; detection power below the 95% confirmatory threshold).

An initial pilot using a manipulation of ±1.5 SD produced pronounced PSOLA artefacts (octave jumps, mechanical quality) across all f_0 conditions. Reducing the manipulation to ±1 SD yielded perceptually natural stimuli; this is therefore an empirical constraint on the manipulation magnitude, not a theoretical preference.

1.2.1 Population norms

The between-speaker population SDs used to calibrate the manipulation targets were derived from Hilton et al.'s (2022) corpus (Hilton et al., 2020). Their GitHub repository provides per-recording acoustic measures (including f_0 mean and f_0 SD) but does not include F3 or F4. To compute the geometric mean of F1–F4 (used as the vocal tract length proxy), F1–F4 were re-extracted from the edited audio files using a custom Praat script.The analysis and all outputs are archived in the population_norms/ folder, available in the Materials component (https://doi.org/10.17605/OSF.IO/H5SNB).

Step 1: Formant extraction. The IDS files (vocalization type `B`, n = 458) were downloaded from Zenodo (Hilton et al., 2020) and filtered to female speakers using the `stimuli-info.csv` metadata file (column `Gender == "F"`). F1–F4 were extracted with the following Praat script[1], using the same Burg analysis settings as Hilton et al. (5500 Hz ceiling, five formants, 25 ms window):

```
# population_norms/extract_formants_IDS.praat
#
# Extract f0 and formant (F1-F4) summary statistics from IDS speech files.
# Outputs one row per file to a CSV.
```

[1] Reproducing this section requires the population_norms/ folder (Materials component, https://doi.org/10.17605/OSF.IO/H5SNB) to be present in the working directory.

```
#
# Adapted from Hilton & Moser et al. (2022) masterscript.praat:
#   https://github.com/themusiclab/infant-speech-song
#   see 'analysis/acoustics_processing/3_masterscript.praat'
#
# Usage:
#   - Run from the Praat Objects window (Open > Run script...)
#   - Point 'directory' at the folder containing the IDS speech .wav files
#     (vocalization type B from the Hilton et al. corpus, e.g. AC001B.wav)
#   - Settings below are for female speakers; adjust pitch/formant bounds
#     for male speakers if needed.

form Extract formant summary stats
    comment Directory of sound files (include trailing slash):
    text directory /path/to/ids_speech_clips/
    sentence Sound_file_extension .wav
    comment Full path of output CSV (include filename and extension):
    text resultsfile /path/to/output/formant_summary.csv
    comment Pitch floor -- use 100 for females, 75 for males
    positive minimum_pitch_(Hz) 100
    comment Pitch ceiling -- use 600 for females, 300 for males
    positive maximum_pitch_(Hz) 600
    comment Maximum formant -- use 5500 for females/children, 5000 for males
    positive maximum_formant_(Hz) 5500
endform

# Build file list
Create Strings as file list... list 'directory$'*'sound_file_extension$'
numberOfFiles = Get number of strings

# Handle existing output file
if fileReadable (resultsfile$)
    pause The file 'resultsfile$' already exists! Overwrite?
    filedelete 'resultsfile$'
endif

# Write CSV header
header$ = "filename,f0_mean,f0_sd,f1_mean,f2_mean,f3_mean,f4_mean'newline$'"
fileappend "'resultsfile$'" 'header$'

# Process each file
for d from 1 to numberOfFiles
    select Strings list
    filename$ = Get string... d
    dotInd = rindex (filename$, ".")
    soundname$ = left$ (filename$, dotInd - 1)
```

```
Read from file... 'directory$''filename$'
soundID = selected ("Sound")

# --- Pitch object (f0 mean and SD) ---
select 'soundID'
To Pitch... 0 'minimum_pitch' 'maximum_pitch'
pitchID = selected ("Pitch")

f0_mean = Get mean... 0 0 Hertz
f0_sd   = Get standard deviation... 0 0 Hertz

# Replace undefined (silent/unvoiced files) with 0
if f0_mean = undefined
    f0_mean = 0
endif
if f0_sd = undefined
    f0_sd = 0
endif

# --- Formant object (Burg, 5 formants) ---
# Settings match Hilton et al.: window 0.025 s, pre-emphasis from 50 Hz
select 'soundID'
To Formant (burg)... 0 5 'maximum_formant' 0.025 50
formantID = selected ("Formant")

select 'formantID'
f1_mean = Get mean... 1 0 0 Hertz
f2_mean = Get mean... 2 0 0 Hertz
f3_mean = Get mean... 3 0 0 Hertz
f4_mean = Get mean... 4 0 0 Hertz

# Replace undefined with 0
if f1_mean = undefined
    f1_mean = 0
endif
if f2_mean = undefined
    f2_mean = 0
endif
if f3_mean = undefined
    f3_mean = 0
endif
if f4_mean = undefined
    f4_mean = 0
endif

# Format to 3 decimal places
f0_mean$ = fixed$ (f0_mean, 3)
```

```
    f0_sd$   = fixed$ (f0_sd,   3)
    f1_mean$ = fixed$ (f1_mean, 3)
    f2_mean$ = fixed$ (f2_mean, 3)
    f3_mean$ = fixed$ (f3_mean, 3)
    f4_mean$ = fixed$ (f4_mean, 3)

    resultline$ = "'soundname$','f0_mean$','f0_sd$',"
    ... "'f1_mean$','f2_mean$','f3_mean$',"
    ... "'f4_mean$''newline$'"

    # Clean up objects for this file before next iteration
    select all
    minus Strings list
    Remove
endfor

select all
Remove

printline Done. Results written to 'resultsfile$'
```

Step 2: Population norm derivation. The extracted formant data were joined with the Hilton et al. acoustic CSV and speaker metadata, filtered to female IDS speakers, and aggregated to one row per speaker before computing between-speaker SDs.

```
# Full analysis code is in population_norms/hilton2022_population_norms.R
# Precomputed outputs loaded from population_norms/by_speaker.csv;
# summary exported to population_norms.csv
readLines("population_norms/hilton2022_population_norms.R") |> cat(sep = "\n")
```

Figure 1 shows the between-speaker distributions for all three parameters. The derived manipulation targets are summarised in Table 2, together with cross-checks from Cox et al. (2023) and Broesch and Bryant (2015).

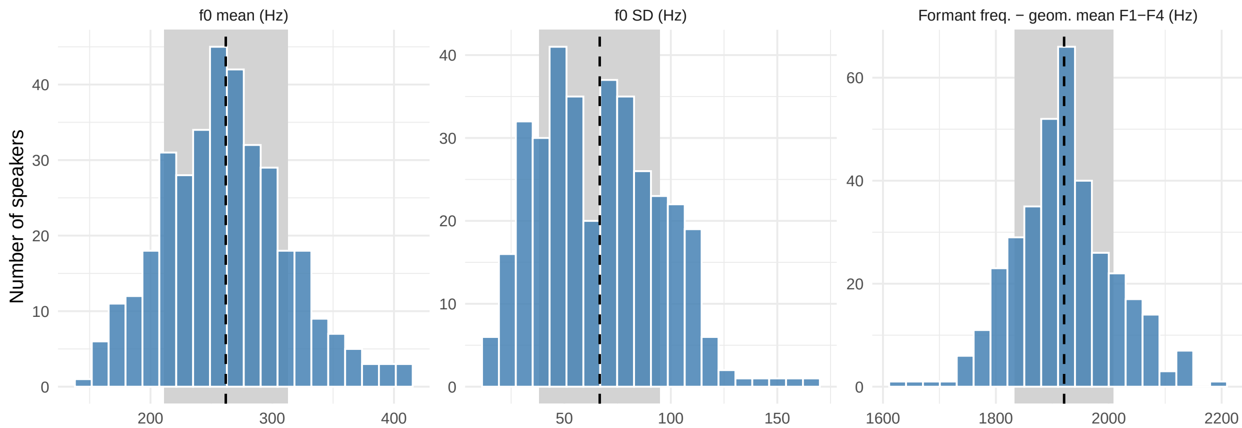

Figure S1. Between-speaker distributions of the three acoustic parameters in female infant-directed speech, derived from Hilton et al. (2022). Vertical dashed lines show population means; shaded bands show ±1 SD (the manipulation targets).

Table S2. Population norms for the three acoustic parameters derived from female IDS speakers in Hilton et al. (2022), and cross-checks from independent sources. All manipulation targets are set at ±1 between-speaker SD. The formant frequencies target is a proportional scaling factor (CV × 1).

Parameter	Pop. mean	Pop. SD	Target (±1 SD)	Cross-check
f_0 mean	261.8 Hz	50.8 Hz	±51 Hz	[a]
f_0 SD	66.6 Hz	28.4 Hz	±28 Hz	[b]
Formant freq.	1920.9 Hz	87.4 Hz	±4.6%	—

[a] Cox et al., 2023, meta-analytic OSF data; n = 202 female speakers; SD = 40.1 Hz.
[b] Broesch and Bryant, 2015, Table 1; Western SD = 24.0 Hz, non-Western SD = 22.4 Hz.

Baseline f_0 mean and f_0 SD are estimated from each recording using Praat's autocorrelation-based pitch analysis; formant frequencies are estimated using Burg's method with a 5500 Hz analysis ceiling, consistent with the synthesis ceiling. Manipulation proceeds in two sequential steps per stimulus.

Step 1 — Formant frequencies. Formant frequencies are manipulated first, applied directly to the original (unmodified) recording, via an LPC source-filter resynthesis pipeline adapted from the Praat Vocal Toolkit (Corretgé, 2012). The manipulation proportionally scales all formant frequencies by the target ratio $r = 1 \pm k_{\text{formant}}$, thereby shifting apparent vocal tract length while leaving the pitch contour unchanged. Briefly: (a) the spectral content above the analysis ceiling (5500 Hz for female speakers) is preserved as a high-frequency residual; (b) the recording is resampled to twice the analysis ceiling; (c) a robust formant analysis is performed and converted to an LPC filter; (d) the signal is inverse-filtered to isolate the glottal source; (e) all formant frequencies in a copy of the formant object are scaled by r; (f) the modified formant object is converted to a new LPC filter and used to re-filter the source; (g) the result is resampled to the original rate and the high-frequency residual is added back. Because this pipeline operates entirely in the spectral domain and does not involve pitch-synchronous resynthesis (PSOLA), the pitch contour of the original recording is left unchanged.

To account for the small discrepancy between the programmed and realised formant shift that arises from the Burg tracker operating on the resynthesised signal, an iterative correction procedure is applied. After each resynthesis, the geometric mean of F1–F4 is measured; if the absolute difference between the measured and target value exceeds 10 Hz, the scaling ratio is updated proportionally as $r \leftarrow r \cdot (\text{target}/\text{measured})$. This repeats up to a maximum of twelve iterations; convergence is typically achieved within two to three iterations.

Step 2 — f_0 mean and f_0 SD. f_0 mean and f_0 SD are manipulated jointly in a second step, applied to the formant-shifted intermediate. An autocorrelation-based pitch estimate is computed from this intermediate, yielding its actual f_0 mean (μ_{src}) and f_0 SD (σ_{src}). An affine transform is then applied to every voiced frame:

$$f_0'(t) = a + b \cdot f_0(t)$$

where the slope $b = \sigma_{\text{target}}/\sigma_{\text{src}}$ scales the pitch distribution width, and the intercept $a = \mu_{\text{target}} - \mu_{\text{src}} \cdot b$ shifts its centre. The modified pitch object is converted to a PitchTier, inserted into a Praat Manipulation object, and the final sound is obtained via overlap-add PSOLA resynthesis.

Because PSOLA resynthesis introduces a small but systematic discrepancy between the programmed and realised pitch SD, arising from differences between the fixed-timestep pitch analysis grid and the pitch-period-based PitchTier representation, an iterative correction procedure is applied. After each resynthesis, the output is re-analysed; if the absolute difference between the measured and target SD exceeds 0.5 Hz, the slope is updated multiplicatively as $b \leftarrow b \cdot (\sigma_{\text{target}}/\sigma_{\text{measured}})$, with the intercept recomputed from the fixed source mean. This repeats up to a maximum of five iterations; convergence is typically achieved within two to three iterations.

Verification. All eight output files are re-analysed independently after saving using the same pitch and formant analysis settings as the baseline (5500 Hz ceiling throughout, consistent with synthesis). The geometric mean of F1–F4 is used as the formant frequency metric, consistent with the population norm derivation. The realised f_0 mean, f_0 SD, and formant frequencies of each stimulus will be reported in the manuscript. The full script is available at https://doi.org/10.17605/OSF.IO/H5SNB.

2 simulation-based power analysis

2.1 Packages

All packages are available from CRAN. The analysis uses the `tidyverse` ecosystem (Wickham et al., 2019) for data manipulation and visualisation, `lmerTest` (Kuznetsova et al., 2017) for LMMs with Satterthwaite degrees of freedom and *p*-values, `emmeans` (Lenth & Piaskowski, 2026) for marginal contrasts and confidence intervals, `effectsize` (Ben-Shachar et al., 2020) for partial ω^2 estimates, `furrr` (Vaughan et al., 2026) for parallel Monte Carlo sampling, `brms` (Bürkner, 2017, 2018) for Bayesian LMMs via Stan, `bayestestR` (Makowski et al., 2019) for ROPE-based equivalence testing, and `posterior` (Bürkner et al., 2026) for summarising and diagnosing posterior distributions. Visualisation uses `ggdist` (Kay, 2024, 2025), `tidyquant` (Dancho & Vaughan, 2026), `ggpubr` (Kassambara, 2026), and `tinytable` (Arel-Bundock, 2026) for publication-quality tables. The Bayesian models are compiled and sampled via `cmdstanr` (Gabry et al., 2025), which provides an R interface to *CmdStan* and is used as the backend for `brms`.

```
library(tidyverse)    # Data manipulation and visualisation
library(lmerTest)     # LMMs with Satterthwaite df and p-values
library(emmeans)      # Marginal contrasts and CIs
library(effectsize)   # Partial omega-squared
library(furrr)        # Parallel Monte Carlo via purrr
library(future)       # Parallel back-end for furrr
library(ggdist)       # Raincloud plot components
library(tidyquant)    # Colour palettes and themes
library(ggpubr)       # Multi-panel figure arrangement
library(scales)       # Axis formatting utilities
library(tinytable)    # Publication-quality tables (PDF-safe)
library(brms)         # Bayesian LMMs via Stan
library(bayestestR)   # ROPE and equivalence testing for Bayesian models
library(posterior)    # Posterior manipulation (brms dependency)
library(cmdstanr)     # Stan backend for brms (faster than rstan)
set_cmdstan_path()    # Locate CmdStan installation (required each session)
if (!dir.exists(cmdstan_path())) {
  stop("CmdStan not found. Run cmdstanr::install_cmdstan() first.")
}
```

2.2 Global Parameters

All key parameters are defined here. Changes propagate automatically throughout the document.

```
#   Simulation scale
set.seed(2025)            # Global seed for reproducibility
n_pop        <- 10000    # Size of each simulated superpopulation
n_sim        <- 1000     # Monte Carlo iterations per sample size (frequentist)
n_sim_bayes  <- 1000     # brms simulations at target n

sample_sizes <- seq(100, 300, by = 10)  # Sample sizes evaluated
target_n     <- 200      # Pre-specified target sample size

#   Analysis parameters
```

```
alpha    <- 0.05 # Significance level (frequentist detection)
ci_level <- 0.90 # CI level for equivalence tests (= 1 - 2*alpha)
eq_bound <- 150  # Equivalence bound (ms); justification in Section 2.3

#   Simulated effect sizes (ms) - H1 population
# Main effects (H1: f0 SD, H2: f0 mean; formant frequencies exploratory)
eff_f0sd    <- 100 # f0 SD: expected mean difference (High vs. Low), ms
eff_f0mean  <-  75 # f0 mean: expected mean difference (High vs. Low), ms
eff_ffreq   <-  50 # Formant freq.: expected mean difference (High vs. Low), ms

# Interaction effects (exploratory only)
eff_f0sd_f0mean  <- 100  # f0 SD x f0 mean
eff_f0sd_ffreq   <- 100  # f0 SD x formant freq.
eff_f0mean_ffreq <-  50  # f0 mean x formant freq.
eff_triple       <- 120  # f0 SD x f0 mean x formant freq. (triple interaction)

#   Data-generating model parameters (shared by both populations)
baseline    <- 2500 # Baseline fixation time (ms; all predictors = Low)
residual_sd <- 1000 # Trial-level residual SD (ms)
y_min       <- 0    # Floor of fixation-time distribution (ms)
y_max       <- 5000 # Ceiling of fixation-time distribution (ms)

#   brms ROPE range (same as frequentist equivalence bounds)
rope_range <- c(-eq_bound, eq_bound)
```

2.3 Equivalence Bounds: Justification and Pre-specification

2.3.1 Rationale for Equivalence Testing

Conventional null-hypothesis significance testing (NHST) cannot distinguish between a true null effect and an underpowered study. For a registered report, it is therefore essential to pre-specify a procedure that permits positive evidence for the absence of a meaningful effect. The two one-sided tests (TOST) procedure achieves this by testing whether an observed effect lies within pre-defined equivalence bounds $[-\Delta, +\Delta]$, or equivalently, whether the $(1-2\alpha)$ CI falls entirely within those bounds (Lakens, 2017).

2.3.2 Defining the Smallest Effect Size of Interest (SESOI)

The equivalence bounds must be anchored to a *smallest effect size of interest* (SESOI): the minimum difference in fixation time that would be theoretically or practically meaningful. The two confirmatory hypotheses are associated with expected effects of 100 ms (f_0 SD, H1) and 75 ms (f_0 mean, H2). The exploratory formant frequencies parameter assumes an expected effect of 50 ms (see Section 2.4). Rather than defining the SESOI purely in terms of these predicted effects, we anchor it to the **precision limit of the design**: the smallest effect that a well-powered infant eye-tracking study of feasible size could realistically detect.

With a residual standard deviation of $\sigma \approx 1000$ ms (consistent with published infant eye-tracking data) and $n = 200$ participants each contributing 4 observations per level of each factor, the approximate standard error of the marginal High − Low contrast is:

$$SE \approx \sqrt{\frac{\sigma^2}{2n}} = \sqrt{\frac{1000^2}{2 \times 200}} \approx 50 \text{ ms}$$

For 80% equivalence power, the required bounds are:

$$\Delta_{80\%} \approx SE \times (z_{0.90} + z_{0.80}) = 50 \times 2.927 \approx 146 \text{ ms}$$

We set equivalence bounds at $\Delta = \pm 150$ ms (the rounded value closest to $\Delta_{80\%}$ that remains scientifically interpretable). An effect smaller than 150 ms represents 3% of the total fixation-time range (0–5000 ms) and 150% of the largest expected main effect (f_0 SD = 100 ms). Such an effect would be undetectable by any practically feasible infant eye-tracking study and is therefore an appropriate threshold for inferring negligibility.

2.3.3 Equivalence Power

The H_0 Monte Carlo simulation in Section 2.5 confirms that, at $n = 200$ with bounds of ± 150 ms, equivalence power reaches approximately 80%. This is an honest reflection of the precision constraints of the infant eye-tracking paradigm: the high trial-level noise ($\sigma \approx 1000$ ms) means that ruling out effects smaller than 150 ms requires the full target sample. Equivalence power at smaller n (e.g., $n = 160$) would be substantially lower (~70%), which is one motivation for the planned sample size to $n = 200$.

2.3.4 Operationalisation

2.3.4.1 Frequentist equivalence test

The marginal pairwise contrast (High − Low, averaged over all levels of the other two factors) is estimated using `emmeans` with Satterthwaite degrees of freedom. A 90% CI is computed. Equivalence is concluded if:

$$\text{Lower CL} > -150 \text{ ms} \;\text{ and }\; \text{Upper CL} < 150 \text{ ms}$$

Equivalence power is estimated from a dedicated H_0 superpopulation in which all acoustic effects are zero (see Section 2.4.2), ensuring the estimate describes the correct conditional probability.

2.3.4.2 Bayesian ROPE

The same bounds define the ROPE for the Bayesian sensitivity analysis. Weakly informative priors place no strong prior mass near zero, so high ROPE percentages reflect genuine posterior evidence for equivalence. Following (John K. Kruschke, 2015), we report both the ROPE percentage and whether the 90% Highest Density Interval (HDI) (John K. Kruschke, 2015; Makowski et al., 2019) falls entirely within the ROPE (strong equivalence).

2.4 Simulated Population Data

2.4.1 H_1 Population: Non-Zero Acoustic Effects

Each participant observes all eight combinations of three acoustic factors, f_0 SD (Low/High), f_0 mean (Low/High), and formant frequencies (Low/High), yielding a 2×2×2 fully crossed within-subjects design. The outcome is total fixation time (ms), bounded to [0, 5000]. We simulate an H_1 superpopulation of $N = 10,000$ participants (×8 conditions = 80,000 rows) from which Monte Carlo samples are drawn to estimate **detection power**.

The data-generating model is:

$$
\begin{aligned}
Y_{ij} = \mu &+ \beta_1 f_0\mathrm{SD}_j + \beta_2 f_0\mathrm{mean}_j + \beta_3 \mathrm{formant}_j + \beta_{12}(f_0\mathrm{SD} \times f_0\mathrm{mean})_j \\
&+ \beta_{13}(f_0\mathrm{SD} \times \mathrm{formant})_j + \beta_{23}(f_0\mathrm{mean} \times \mathrm{formant})_j + \beta_{123}(f_0\mathrm{SD} \times f_0\mathrm{mean} \times \mathrm{formant})_j + \epsilon_{ij}
\end{aligned}
$$

where $\mu = 2500$ ms, the β coefficients correspond to the effect sizes in Section 2.2, and $\epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$, truncated to $[0, 5000]$.

```r
# Helper: build a fully crossed superpopulation --------------------------------
make_population <- function(n, seed_offset,
                            b_f0sd, b_f0mean, b_ffreq,
                            b_f0sd_f0mean, b_f0sd_ffreq,
                            b_f0mean_ffreq, b_triple,
                            mu    = baseline,
                            sigma = residual_sd,
                            ymin  = y_min,
                            ymax  = y_max) {

  set.seed(2025 + seed_offset)

  conditions <- expand_grid(
    f0_sd   = c("Low", "High"),
    f0_mean = c("Low", "High"),
    ffreq   = c("Low", "High")
  )

  ids    <- str_c("P", str_pad(seq_len(n), width = nchar(n), pad = "0"))
  design <- expand_grid(ID = ids, conditions)

  design |>
    mutate(
      f0_sd_v   = if_else(f0_sd   == "High", 1L, 0L),
      f0_mean_v = if_else(f0_mean == "High", 1L, 0L),
      ffreq_v   = if_else(ffreq   == "High", 1L, 0L),
      mu_i =  mu
              + b_f0sd         * f0_sd_v
              + b_f0mean       * f0_mean_v
              + b_ffreq        * ffreq_v
```

```
              + b_f0sd_f0mean  * f0_sd_v  * f0_mean_v
              + b_f0sd_ffreq   * f0_sd_v  * ffreq_v
              + b_f0mean_ffreq * f0_mean_v * ffreq_v
              + b_triple       * f0_sd_v  * f0_mean_v * ffreq_v,
      fixation_time = round(rnorm(n(), mean = mu_i, sd = sigma)),
      fixation_time = pmin(pmax(fixation_time, ymin), ymax)
    ) |>
    mutate(
      f0_sd   = factor(f0_sd,   levels = c("Low", "High")),
      f0_mean = factor(f0_mean, levels = c("Low", "High")),
      ffreq   = factor(ffreq,   levels = c("Low", "High"))
    ) |>
    select(ID, f0_sd, f0_mean, ffreq, f0_sd_v, f0_mean_v, ffreq_v, fixation_time)
}
```

```
# H1 superpopulation (non-zero effects; used for detection power) --------------
h1_data <- make_population(
  n              = n_pop,
  seed_offset    = 0,
  b_f0sd         = eff_f0sd,
  b_f0mean       = eff_f0mean,
  b_ffreq        = eff_ffreq,
  b_f0sd_f0mean  = eff_f0sd_f0mean,
  b_f0sd_ffreq   = eff_f0sd_ffreq,
  b_f0mean_ffreq = eff_f0mean_ffreq,
  b_triple       = eff_triple
)
```

2.4.2 H_0 Population: All Acoustic Effects Zero

Detection power and equivalence power must be estimated from **separate** superpopulations. Detection power asks: given real effects, how often do we detect them? Equivalence power asks: given no effects, how often do we correctly conclude equivalence? The H_0 superpopulation is structurally identical to H_1 but with all acoustic effects set to zero.

```
# H0 superpopulation (all acoustic effects = 0; used for equivalence power) ---
h0_data <- make_population(
  n              = n_pop,
  seed_offset    = 1,   # different seed -> independent population
  b_f0sd         = 0,
  b_f0mean       = 0,
  b_ffreq        = 0,
  b_f0sd_f0mean  = 0,
  b_f0sd_ffreq   = 0,
  b_f0mean_ffreq = 0,
  b_triple       = 0
)
```

2.4.3 Population-Level Model and Effect Sizes

```r
# Intercept-only random effects: appropriate at population scale and
# consistent with the signal-to-noise ratio of this design (see main text)
pop_mod <- lmer(
  fixation_time ~ f0_sd * f0_mean * ffreq + (1 | ID),
  data = h1_data
)

pop_omega <- omega_squared(pop_mod, partial = TRUE) |>
  rename(Effect = Parameter)
```

```r
anova(pop_mod) |>
  as.data.frame() |>
  rownames_to_column(var = "Effect") |>
  left_join(pop_omega |> select(Effect, Omega2_partial), by = "Effect") |>
  mutate(
    Hypothesis = case_when(
      Effect == "f0_sd"   ~ "H1",
      Effect == "f0_mean" ~ "H2",
      Effect == "ffreq"   ~ "Exploratory",
      TRUE                ~ "Exploratory"
    ),
    Effect = str_replace_all(Effect, "f0_sd",   "$f_0$ SD"),
    Effect = str_replace_all(Effect, "f0_mean", "$f_0$ mean"),
    Effect = str_replace_all(Effect, "ffreq",   "Formant freq."),
    Effect = str_replace_all(Effect, ":",       " $\\times$ "),
    DenDF          = round(DenDF, 1),
    `F value`      = round(`F value`, 2),
    Omega2_partial = round(Omega2_partial, 4)
  ) |>
  select(Hypothesis, Effect, `F value`, NumDF, DenDF, Omega2_partial) |>
  unite("$df$", NumDF:DenDF, sep = ", ") |>
  rename(
    "Hypothesis"    = Hypothesis,
    "Fixed effect"  = Effect,
    "$F$"           = `F value`,
    "$\\omega^2_p$" = Omega2_partial
  ) |>
  tt(notes = "\\textit{Note.} Partial $\\omega^2_p$ benchmarks \\cite{field2013}:
     very small $< 0.01$; small $= 0.01$--$0.06$; medium $= 0.06$--$0.14$;
     large $\\geq 0.14$.") |>
  style_tt(align = "llccc") |>
  format_tt(escape = FALSE)
```

Table S3. Population-level fixed effects from the LMM fitted to the full H_1 superpopulation (N = 10,000). ANOVA table with Satterthwaite degrees of freedom, supplemented by partial ω^2. Confirmatory effects (H1–H2) in the first two rows; formant frequencies (exploratory) in the third row; exploratory interactions follow.

Hypothesis	Fixed effect	F	df	ω_p^2
H1	f_0 SD	1161.19	1, 69993	0.0163
H2	f_0 mean	646.48	1, 69993	0.0091
Exploratory	Formant freq.	443.99	1, 69993	0.0063
Exploratory	f_0 SD *times* f_0 mean	135.26	1, 69993	0.0019
Exploratory	f_0 SD *times* Formant freq.	126.19	1, 69993	0.0018
Exploratory	f_0 mean *times* Formant freq.	56.12	1, 69993	0.0008
Exploratory	f_0 SD *times* f_0 mean *times* Formant freq.	13.92	1, 69993	0.0002

Note. Partial ω_p^2 benchmarks Field, 2013: very small < 0.01; small $= 0.01$–0.06; medium $= 0.06$–0.14; large ≥ 0.14.

2.4.4 Distribution of Fixation Times Across Conditions (H_1 Population)

```r
set.seed(1)
plot_data <- h1_data |> slice_sample(prop = 0.02)

label_f0mean <- c(
  "Low"  = "paste('Low ',  italic(f)[0], ' mean')",
  "High" = "paste('High ', italic(f)[0], ' mean')"
)
label_ffreq <- c(
  "Low"  = "'Low formant freq.'",
  "High" = "'High formant freq.'"
)

ggplot(h1_data, aes(x = f0_sd, y = fixation_time, fill = f0_sd)) +
  stat_halfeye(adjust = 0.5, justification = -0.3,
               .width = 0, point_colour = NA, alpha = 0.65) +
  geom_jitter(data = plot_data, aes(colour = f0_sd),
              width = 0.1, alpha = 0.3, size = 0.5) +
  stat_summary(fun = mean, geom = "point", size = 2.5, colour = "black",
               position = position_nudge(x = 0.22)) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.1,
               colour = "black", position = position_nudge(x = 0.22)) +
  facet_grid(
    ffreq ~ f0_mean,
    labeller = labeller(
      f0_mean = as_labeller(label_f0mean, default = label_parsed),
      ffreq   = as_labeller(label_ffreq,  default = label_parsed)
    )
```

```
  ) +
  scale_fill_tq() +
  scale_colour_tq() +
  labs(
    x      = expression(italic(f)[0] ~ "SD"),
    y      = "Total fixation time (ms)",
    fill   = expression(italic(f)[0] ~ "SD"),
    colour = expression(italic(f)[0] ~ "SD")
  ) +
  theme_tq(base_size = 12) +
  theme(legend.position = "bottom")
```

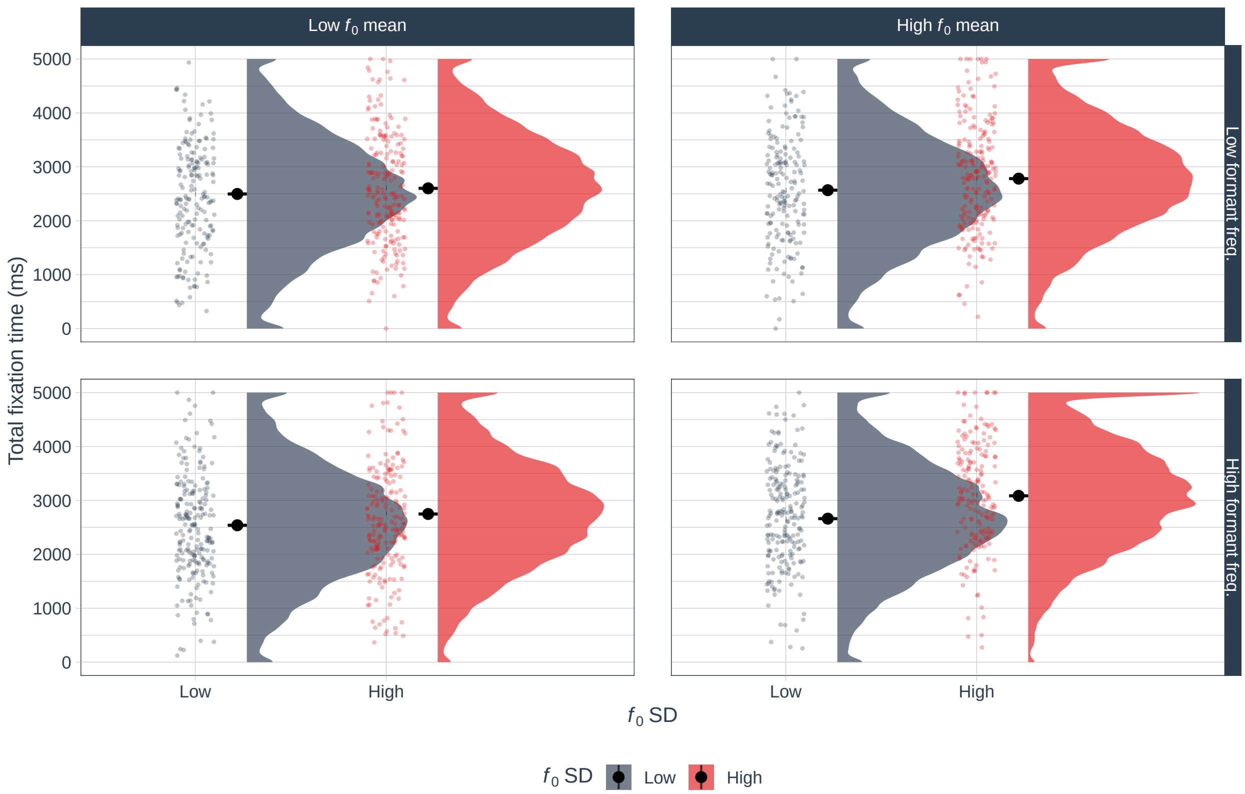

Figure S2. Distribution of fixation times in the H_1 superpopulation as a function of f_0 SD (x-axis), f_0 mean (columns), and formant frequencies (rows). Each panel shows the density half-violin, jittered data points (2% random sample), and condition mean ± SE. Fixation times increase systematically with f_0 SD (the largest main effect), with additive contributions from f_0 mean and formant frequencies.

2.5 Monte Carlo Power Analysis: Frequentist

2.5.1 Simulation Strategy

Detection and equivalence power are estimated from two independent Monte Carlo simulations:

- **Detection power** (H_1 population): 1000 samples per candidate n from `h1_data`. For each sample, the LMM ANOVA p-value is recorded for each main effect. Power = proportion of samples with $p < 0.05$.
- **Equivalence power** (H_0 population): 1000 samples per candidate n from `h0_data`. For each sample, the marginal High − Low contrast is estimated via `emmeans` (Satterthwaite df) with a 90% CI. Power = proportion of samples where the CI falls entirely within [-150, 150] ms.

Both loops cover n = 100 to 300 (steps of 10) and run in parallel via `furrr`.

2.5.2 Simulation Functions

```r
# Detection: p-values from LMM ANOVA (H1 data) --------------------------------
run_detection_sim <- function(dat, n_sample, n_sim, alpha = 0.05) {
  map_dfr(seq_len(n_sim), function(i) {
    ids  <- dat |> distinct(ID) |> slice_sample(n = n_sample) |> pull(ID)
    sdat <- dat |> filter(ID %in% ids)
    mod  <- suppressMessages(
      lmer(fixation_time ~ f0_sd * f0_mean * ffreq + (1 | ID), data = sdat)
    )
    av    <- anova(mod)
    terms <- c("f0_sd", "f0_mean", "ffreq")
    map_dfr(terms, function(trm) {
      tibble(
        sim      = i,
        n_sample = n_sample,
        term     = trm,
        p_detect = av[trm, "Pr(>F)"],
        detect   = av[trm, "Pr(>F)"] < alpha
      )
    })
  })
}

# Equivalence: 90% CIs from emmeans, check ROPE containment (H0 data) ---------
run_equivalence_sim <- function(dat, n_sample, n_sim,
                                ci_lev = 0.90, bound = eq_bound) {
  map_dfr(seq_len(n_sim), function(i) {
    ids  <- dat |> distinct(ID) |> slice_sample(n = n_sample) |> pull(ID)
    sdat <- dat |> filter(ID %in% ids)
    mod  <- suppressMessages(
      lmer(fixation_time ~ f0_sd * f0_mean * ffreq + (1 | ID), data = sdat)
    )
    terms <- c("f0_sd", "f0_mean", "ffreq")
```

```
    map_dfr(terms, function(trm) {
      em <- emmeans(mod, as.formula(paste0("~ ", trm)))
      ct <- pairs(em, reverse = TRUE)
      ci <- confint(ct, level = ci_lev)
      tibble(
        sim      = i,
        n_sample = n_sample,
        term     = trm,
        estimate = ci$estimate[1],
        ci_lo    = ci$lower.CL[1],
        ci_hi    = ci$upper.CL[1],
        sig      = ci$lower.CL[1] > 0 | ci$upper.CL[1] < 0,
        equiv    = ci$lower.CL[1] > -bound & ci$upper.CL[1] < bound
      )
    })
  })
}
```

2.5.3 Run Detection Simulation (H_1 Population)

```
plan(multisession)

detect_results <- map_dfr(
  sample_sizes,
  ~ run_detection_sim(
      dat      = h1_data,
      n_sample = .x,
      n_sim    = n_sim,
      alpha    = alpha
    )
)

plan(sequential)
```

2.5.4 Run Equivalence Simulation (H_0 Population)

```
plan(multisession)

equiv_results <- map_dfr(
  sample_sizes,
  ~ run_equivalence_sim(
      dat      = h0_data,
      n_sample = .x,
      n_sim    = n_sim,
      ci_lev   = ci_level,
      bound    = eq_bound
    )
```

```
)

plan(sequential)
```

2.5.5 Summarise Power

```
term_labs <- c(
  "f0_sd"   = "italic(f)[0]~SD",
  "f0_mean" = "italic(f)[0]~mean",
  "ffreq"   = "Formant~freq."
)

detect_summ <- detect_results |>
  group_by(term, n_sample) |>
  summarise(power_detect = mean(detect, na.rm = TRUE), .groups = "drop")

equiv_summ <- equiv_results |>
  group_by(term, n_sample) |>
  summarise(power_equiv = mean(equiv, na.rm = TRUE), .groups = "drop")

power_summ <- detect_summ |>
  left_join(equiv_summ, by = c("term", "n_sample")) |>
  mutate(term = factor(term, levels = names(term_labs), labels = term_labs))

# Minimum n achieving > 95% detection power per effect
n_80_detect <- detect_summ |>
  filter(power_detect > 0.95) |>
  group_by(term) |>
  slice_min(n_sample) |>
  select(term, n_80_det = n_sample, pwr_det_80 = power_detect)

n_recommended       <- max(n_80_detect$n_80_det)
detect_at_target    <- detect_summ |> filter(n_sample == target_n)
equiv_at_target     <- equiv_summ  |> filter(n_sample == target_n)
```

2.5.6 Power Curves

```
power_threshold_detect <- 0.95
power_threshold_equiv  <- 0.80

make_power_plot <- function(data, y_var, title_str, threshold) {
  ggplot(data, aes(x = n_sample, y = .data[[y_var]], group = term)) +
    geom_line(linewidth = 0.5, colour = "grey40") +
    geom_point(aes(colour = .data[[y_var]] > threshold), size = 2) +
    geom_hline(yintercept = threshold,
               linetype = "dashed", colour = "red", linewidth = 0.6) +
    geom_vline(xintercept = target_n,
```

```
               linetype = "dotted", colour = "#2c3e50", linewidth = 0.6) +
    annotate("text", x = target_n + 2, y = 0.12,
             label = paste0("n = ", target_n),
             hjust = 0, colour = "#2c3e50", size = 3) +
    facet_wrap(
      ~ term, nrow = 1,
      labeller = as_labeller(
        setNames(levels(data$term), levels(data$term)),
        default = label_parsed
      )
    ) +
    scale_colour_tq(
      labels = c("FALSE" = "At or below target", "TRUE" = "Above target"),
      name   = "Power"
    ) +
    scale_y_continuous(labels = percent_format(accuracy = 1), limits = c(0, 1)) +
    labs(title = title_str, x = "Sample size (n)", y = "Estimated power") +
    theme_tq(base_size = 11) +
    theme(legend.position = "bottom")
}

ggarrange(
  make_power_plot(
    power_summ, "power_detect",
    "A  Detection power (frequentist, p < .05) \u2014 H\u2081 population",
    threshold = power_threshold_detect
  ),
  make_power_plot(
    power_summ, "power_equiv",
    paste0("B  Equivalence power (90% CI within \u00b1",
           eq_bound, " ms) \u2014 H\u2080 population"),
    threshold = power_threshold_equiv
  ),
  nrow = 2, common.legend = TRUE, legend = "bottom"
)
```

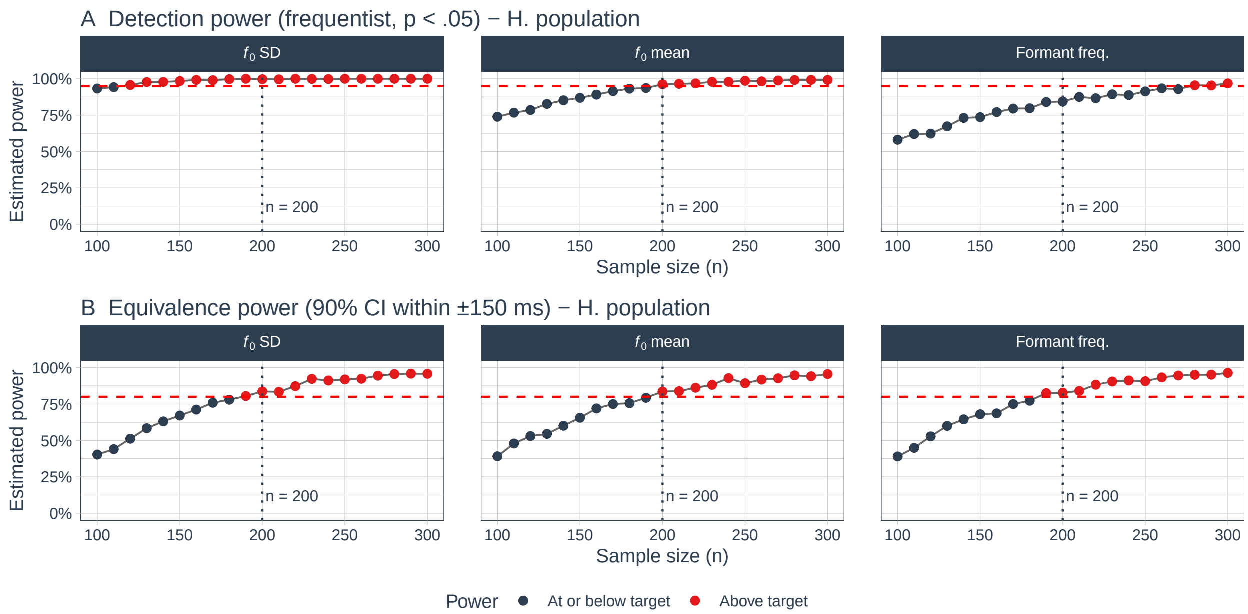

Figure S3. Monte Carlo power curves for H1 (f_0 SD), H2 (f_0 mean), and formant frequencies (exploratory; shown for transparency). **Panel A:** Frequentist detection power ($p < .05$ in the LMM ANOVA), estimated from the H_1 superpopulation. The dashed line marks the 95% target; this threshold applies only to the confirmatory hypotheses (H1 and H2) — no power constraint applies to the exploratory formant frequencies parameter. **Panel B:** Frequentist equivalence power (90% CI within ±150 ms), estimated from the H_0 superpopulation (all acoustic effects = 0). The dashed line marks the 80% equivalence power target (see Section 2.3). The dotted vertical line marks $n = 200$. Points are coloured by whether power is above or at/below the panel-specific target.

2.5.7 Distribution of Contrast Estimates at Target Sample Size

```
term_factor <- function(x) {
  factor(x, levels = names(term_labs), labels = term_labs)
}

# Use equivalence simulation output for H1 (already has estimate + CI)
set.seed(42)
h1_ci_sample <- run_equivalence_sim(
  dat       = h1_data,
  n_sample = target_n,
  n_sim     = 200,        # lighter subsample for plotting only
  ci_lev    = ci_level,
  bound     = eq_bound
) |> mutate(term_lab = term_factor(term))

h0_at_target <- equiv_results |>
  filter(n_sample == target_n) |>
  mutate(term_lab = term_factor(term))
```

```
make_dist_plot <- function(dat, title_str) {
  dat <- dat |>
    mutate(
      outcome = case_when(
        equiv            ~ "Equivalent",
        sig & !equiv     ~ "Significant, not equivalent",
        TRUE             ~ "Inconclusive"
      ),
      outcome = factor(outcome,
                       levels = c("Significant, not equivalent",
                                  "Equivalent",
                                  "Inconclusive"))
    )
  ggplot(dat, aes(x = estimate, fill = outcome)) +
    annotate("rect", xmin = -eq_bound, xmax = eq_bound,
             ymin = 0, ymax = Inf, fill = "grey85", alpha = 0.5) +
    geom_vline(xintercept = c(-eq_bound, eq_bound),
               linetype = "dashed", colour = "grey40", linewidth = 0.6) +
    geom_histogram(binwidth = 10, alpha = 0.85, colour = NA) +
    annotate("text", x = 0, y = Inf, vjust = 1.5,
             label = paste0("\u00b1", eq_bound, " ms"),
             fontface = "italic", colour = "grey35", size = 3) +
    facet_wrap(
      ~ term_lab, nrow = 1,
      labeller = as_labeller(
        setNames(levels(dat$term_lab), levels(dat$term_lab)),
        default = label_parsed
      )
    ) +
    scale_fill_manual(
      values = c(
        "Significant, not equivalent" = "#2ecc71",
        "Equivalent"                  = "#D55E00",
        "Inconclusive"                = "grey65"
      ),
      name = NULL
    ) +
    scale_x_continuous(breaks = pretty_breaks(n = 6)) +
    labs(title = title_str,
         x = "Estimated contrast: High \u2212 Low (ms)",
         y = "Frequency") +
    theme_tq(base_size = 11) +
    theme(legend.position = "bottom")
}

ggarrange(
  make_dist_plot(h1_ci_sample,
```

```
                 "A  H\u2081 population (real effects present)"),
  make_dist_plot(h0_at_target,
                 "B  H\u2080 population (all effects = 0)"),
  nrow = 2, common.legend = TRUE, legend = "bottom"
)
```

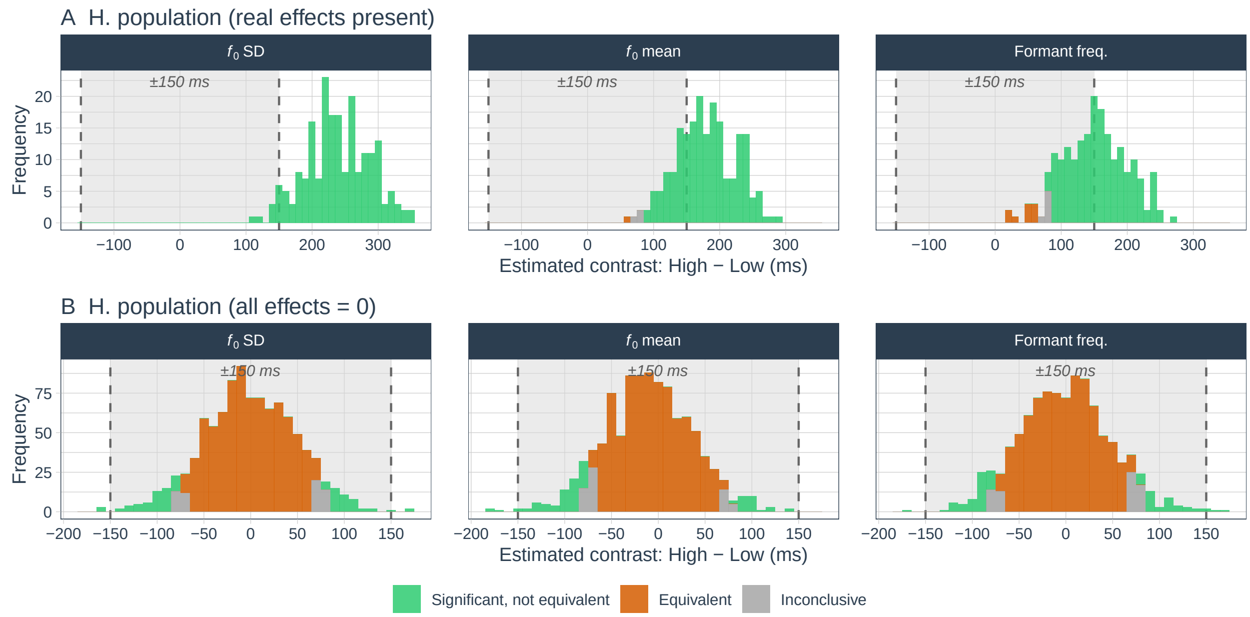

Figure S4. Sampling distributions of the marginal High − Low contrast (ms) across 1000 Monte Carlo samples at $n = 200$. **Panel A ($\mathbf{H_1}$ population):** Under real effects. Distributions are centred near the true effects; the grey band marks the equivalence bounds (±150 ms). Filled bars show the small fraction of samples in which the CI nevertheless fell within the bounds. **Panel B ($\mathbf{H_0}$ population):** Under a true null. Distributions are centred near zero. Filled (orange) bars show simulations where the 90% CI falls entirely within ±150 ms (correct equivalence); grey bars are inconclusive.

2.5.8 Sample Size Summary

```
detect_at_target |>
  left_join(equiv_at_target, by = c("term", "n_sample")) |>
  left_join(n_80_detect,     by = "term") |>
  arrange(factor(term, levels = c("f0_sd", "f0_mean", "ffreq"))) |>
  mutate(
    Effect = case_when(
      term == "f0_sd"   ~ "$f_0$ SD (H1)",
      term == "f0_mean" ~ "$f_0$ mean (H2)",
      term == "ffreq"   ~ "Formant freq. (Exploratory)"
    ),
    `Expected effect` = case_when(
      term == "f0_sd"   ~ paste0(eff_f0sd,    " ms"),
      term == "f0_mean" ~ paste0(eff_f0mean,  " ms"),
```

```
    term == "ffreq"    ~ paste0(eff_ffreq,    " ms")
  ),
  `Detection power`  = paste0(round(power_detect * 100, 1), "\\%"),
  `Equiv. power`     = paste0(round(power_equiv  * 100, 1), "\\%"),
  `Min $n$ (95\\% det.)` = n_80_det
) |>
select(Effect, `Expected effect`, `Detection power`,
       `Min $n$ (95\\% det.)`, `Equiv. power`) |>
tt(notes = paste0(
     "\\textit{Note.} Detection power: $p < .05$, from H$_1$ simulations at
     $n = ", target_n, "$. Target: $\\geq$95\\% for confirmatory hypotheses ",
     "(H1 and H2); formant frequencies are exploratory (no power constraint). ",
     "Equivalence power: 90\\% CI within $\\pm ",
     eq_bound, "$ ms, from H$_0$ simulations at $n = ", target_n,
     "$. Both based on ", n_sim,
     " Monte Carlo samples per candidate $n$."),
   width = c(0.30, 0.18, 0.18, 0.18, 0.16)) |>
style_tt(align = "lcccc") |>
format_tt(escape = FALSE)
```

Table S4. Power summary at the target sample size ($n = 200$). Detection power is from H_1 simulations; equivalence power from H_0 simulations. Equivalence power is intentionally modest given the high trial-level noise of this design (see Section 2.3 for full discussion).

Effect	Expected effect	Detection power	Min n (95% det.)	Equiv. power
f_0 SD (H1)	100 ms	99.7%	120	83.7%
f_0 mean (H2)	75 ms	96.2%	200	83.6%
Formant freq. (Exploratory)	50 ms	84.4%	280	82.8%

Note. Detection power: $p < .05$, from H_1 simulations at $n = 200$. Target: $\geq$95% for confirmatory hypotheses (H1 and H2); formant frequencies are exploratory (no power constraint). Equivalence power: 90% CI within ± 150 ms, from H_0 simulations at $n = 200$. Both based on 1000 Monte Carlo samples per candidate n.

The simulation confirms that a sample of $n = \mathbf{280}$ achieves > 95% detection power for both confirmatory main effects (H1: f_0 SD; H2: f_0 mean), and that the pre-specified target of $n = 200$ further ensures approximately 80% equivalence power at bounds of ± 150 ms (see Section 2.3). Formant frequencies are treated as exploratory, so no detection power threshold applies to that parameter. Both detection and equivalence considerations therefore motivate the target of $n = 200$.

2.6 Bayesian Sensitivity Analysis: brms + ROPE

2.6.1 Rationale and Scope

The Bayesian analysis is a secondary robustness check. Rather than Bayes factors, which require bridge sampling and can be numerically unstable for complex LMMs, we apply the ROPE procedure from `bayestestR` (Makowski et al., 2019) to posteriors from `brms` models. Two sets of 1000 models are fitted at $n = 200$, mirroring the frequentist structure:

- **H_1 models**: data from `h1_data`. Posteriors should lie far from the ROPE, confirming sensitivity.
- **H_0 models**: data from `h0_data`. ROPE percentages should be high, quantifying Bayesian equivalence power.

2.7 Prior Specification

We use **sum contrast coding** (Low $= -0.5$; High $= +0.5$) so that each main-effect coefficient equals the marginal High − Low difference, directly comparable to the `emmeans` marginal contrast. Priors are weakly informative:

```
tibble(
  `Parameter class` = c(
    "Intercept",
    "Main effects and interactions",
    "Residual SD ($\\sigma$)",
    "Random-effect SD"
  ),
  Prior = c(
    "$\\mathcal{N}(2500,\\; 500^2)$",
    "$\\mathcal{N}(0,\\; 200^2)$",
    "Half-$\\mathcal{N}(0,\\; 500^2)$",
    "Half-$\\mathcal{N}(0,\\; 500^2)$"
  ),
  Rationale = c(
    "Grand mean; $\\pm$1 SD spans plausible range",
    "Allows effects up to $\\pm$400 ms; flat over realistic range",
    "Consistent with $\\sigma \\approx 1000$ ms in the DGM",
    "Weakly informative for participant variability"
  )
) |>
  tt(align = "lll", width = c(0.25, 0.30, 0.40)) |>
  format_tt(escape = FALSE)
```

Table S5. Prior specification for Bayesian LMMs.

Parameter class	Prior	Rationale
Intercept	$\mathcal{N}(2500,\ 500^2)$	Grand mean; ±1 SD spans plausible range
Main effects and interactions	$\mathcal{N}(0,\ 200^2)$	Allows effects up to ±400 ms; flat over realistic range
Residual SD (σ)	Half-$\mathcal{N}(0,\ 500^2)$	Consistent with $\sigma \approx 1000$ ms in the DGM
Random-effect SD	Half-$\mathcal{N}(0,\ 500^2)$	Weakly informative for participant variability

```
brms_priors <- c(
  prior(normal(2500, 500), class = Intercept),
  prior(normal(0, 200),    class = b),
  prior(normal(0, 500),    class = sigma),
  prior(normal(0, 500),    class = sd)
)
```

2.7.1 Simulation Function

Reproducibility note. Each model uses seed `base_seed + i` to ensure independent, reproducible chains across simulations.

```
run_bayes_sim <- function(dat, n_sample, n_sim_b, priors,
                          chains = 2, iter = 2000, warmup = 1000,
                          bound = eq_bound, ci_lev = ci_level,
                          base_seed = 9999) {

  map_dfr(seq_len(n_sim_b), function(i) {

    ids  <- dat |> distinct(ID) |> slice_sample(n = n_sample) |> pull(ID)
    sdat <- dat |> filter(ID %in% ids) |>
      mutate(
        f0_sd_c   = if_else(f0_sd   == "High", 0.5, -0.5),
        f0_mean_c = if_else(f0_mean == "High", 0.5, -0.5),
        ffreq_c   = if_else(ffreq   == "High", 0.5, -0.5)
      )

    fit <- brm(
      fixation_time ~ f0_sd_c * f0_mean_c * ffreq_c + (1 | ID),
      data     = sdat,
      family   = gaussian(),
      prior    = priors,
      chains   = chains,
      iter     = iter,
      warmup   = warmup,
```

```
      cores   = chains,
      seed    = base_seed + i,
      refresh = 0,
      silent  = 2,
      backend = "cmdstanr"   # change to "rstan" if cmdstanr is not installed
    )

    eq_res <- equivalence_test(
      fit,
      range = c(-bound, bound),
      ci    = ci_lev
    ) |>
      filter(Parameter %in% c("b_f0_sd_c", "b_f0_mean_c", "b_ffreq_c"))

    post <- as_draws_df(fit)

    tibble(
      sim            = i,
      n_sample       = n_sample,
      term           = c("f0_sd", "f0_mean", "ffreq"),
      posterior_mean = c(mean(post$b_f0_sd_c),
                         mean(post$b_f0_mean_c),
                         mean(post$b_ffreq_c)),
      posterior_sd   = c(sd(post$b_f0_sd_c),
                         sd(post$b_f0_mean_c),
                         sd(post$b_ffreq_c)),
      rope_pct       = eq_res$ROPE_Percentage,
      rope_equiv     = eq_res$ROPE_Equivalence,
      hdi_low        = eq_res$HDI_low,
      hdi_high       = eq_res$HDI_high
    )
  })
}
```

2.7.2 Run Bayesian Simulations

```
bayes_h1 <- run_bayes_sim(
  dat       = h1_data,
  n_sample  = target_n,
  n_sim_b   = n_sim_bayes,
  priors    = brms_priors,
  base_seed = 1000
)
```

```
bayes_h0 <- run_bayes_sim(
  dat       = h0_data,
  n_sample  = target_n,
```

```
  n_sim_b    = n_sim_bayes,
  priors     = brms_priors,
  base_seed = 2000
)
```

2.7.3 Results: ROPE Percentages and Posterior Summaries

```
format_bayes_row <- function(dat, pop_label) {
  dat |>
    group_by(term) |>
    summarise(
      Population = pop_label,
      Effect = case_when(
        first(term) == "f0_sd"   ~ "$f_0$ SD (H1)",
        first(term) == "f0_mean" ~ "$f_0$ mean (H2)",
        first(term) == "ffreq"   ~ "Formant freq. (Exploratory)"
      ),
      `Post. mean (SD)` = paste0(
        round(mean(posterior_mean), 1), " (",
        round(mean(posterior_sd),   1), ")"
      ),
      `Mean ROPE \\%` = paste0(
        round(mean(rope_pct, na.rm = TRUE) * 100, 1), "\\%"
      ),
      `Prop. HDI in ROPE` = paste0(
        round(mean(rope_equiv == "Accepted", na.rm = TRUE) * 100, 1), "\\%"
      ),
      .groups = "drop"
    ) |>
    select(-term)
}

bind_rows(
  format_bayes_row(bayes_h1, "H$_1$ (effects present)"),
  format_bayes_row(bayes_h0, "H$_0$ (effects = 0)")
) |>
  arrange(Effect, Population) |>
  tt(notes = paste0("\\textit{Note.} ROPE $= [\\pm ", eq_bound, "$ ms]. ",
       "ROPE \\% = proportion of posterior within ROPE. ",
       "HDI in ROPE = entire ", ci_level * 100,
       "\\% HDI within ROPE (strong equivalence). ",
       "Based on $n_{\\text{sim}} = ", n_sim_bayes,
       "$ simulations."),
     width = c(0.22, 0.24, 0.20, 0.17, 0.17)) |>
  style_tt(align = "llccc") |>
  format_tt(escape = FALSE)
```

Table S6. Bayesian ROPE analysis across 1000 simulations per population at $n = 200$. Under H_1 (real effects), ROPE percentages should be low (posteriors outside the ROPE). Under H_0 (null effects), ROPE percentages should be high (posteriors within the ROPE).

Population	Effect	Post. mean (SD)	Mean ROPE %	Prop. HDI in ROPE
H_0 (effects = 0)	f_0 SD (H1)	-1.9 (47.8)	98.9%	87.7%
H_1 (effects present)	f_0 SD (H1)	222.9 (47.7)	11.2%	0.1%
H_0 (effects = 0)	f_0 mean (H2)	-9.9 (47.8)	99%	88.4%
H_1 (effects present)	f_0 mean (H2)	167.8 (47.7)	38.4%	2%
H_0 (effects = 0)	Formant freq. (Exploratory)	0.7 (47.9)	99%	87.5%
H_1 (effects present)	Formant freq. (Exploratory)	137.6 (47.8)	57.8%	8%

Note. ROPE = [±150 ms]. ROPE % = proportion of posterior within ROPE. HDI in ROPE = entire 90% HDI within ROPE (strong equivalence). Based on $n_{\text{sim}} = 1000$ simulations.

2.7.4 Posterior Distributions and ROPE Percentages

```r
prep_bayes <- function(dat, label) {
  dat |> mutate(
    Population = label,
    term_lab   = factor(term, levels = names(term_labs), labels = term_labs)
  )
}

bayes_combined <- bind_rows(
  prep_bayes(bayes_h1, "H\u2081 (effects present)"),
  prep_bayes(bayes_h0, "H\u2080 (effects = 0)")
)

p_post <- ggplot(bayes_combined,
                 aes(x = posterior_mean, fill = Population)) +
  annotate("rect", xmin = -eq_bound, xmax = eq_bound,
           ymin = 0, ymax = Inf, fill = "grey85", alpha = 0.5) +
  geom_vline(xintercept = c(-eq_bound, eq_bound),
             linetype = "dashed", colour = "grey40") +
  geom_histogram(bins = 15, colour = "white", alpha = 0.8,
                 position = "identity") +
  facet_wrap(
    ~ term_lab, nrow = 3,
    labeller = as_labeller(
      setNames(levels(bayes_combined$term_lab),
```

```
               levels(bayes_combined$term_lab)),
      default = label_parsed
    ),
    scales = "free_x"
  ) +
  scale_fill_tq() +
  labs(title = "Posterior mean estimates",
       x = "Posterior mean: High \u2212 Low (ms)",
       y = "Count", fill = NULL) +
  theme_tq(base_size = 11) +
  theme(legend.position = "bottom")

p_rope_pct <- ggplot(bayes_combined,
                     aes(x = rope_pct * 100, fill = Population)) +
  geom_histogram(bins = 15, colour = "white", alpha = 0.8,
                 position = "identity") +
  geom_vline(xintercept = 50, linetype = "dashed", colour = "red") +
  facet_wrap(
    ~ term_lab, nrow = 3,
    labeller = as_labeller(
      setNames(levels(bayes_combined$term_lab),
               levels(bayes_combined$term_lab)),
      default = label_parsed
    )
  ) +
  scale_fill_tq() +
  scale_x_continuous(labels = function(x) paste0(x, "%")) +
  labs(title = "ROPE percentages",
       x = paste0("% of posterior within \u00b1", eq_bound, " ms"),
       y = "Count", fill = NULL) +
  theme_tq(base_size = 11) +
  theme(legend.position = "bottom")

ggarrange(p_post, p_rope_pct,
          labels = "AUTO", ncol = 2,
          common.legend = TRUE, legend = "bottom")
```

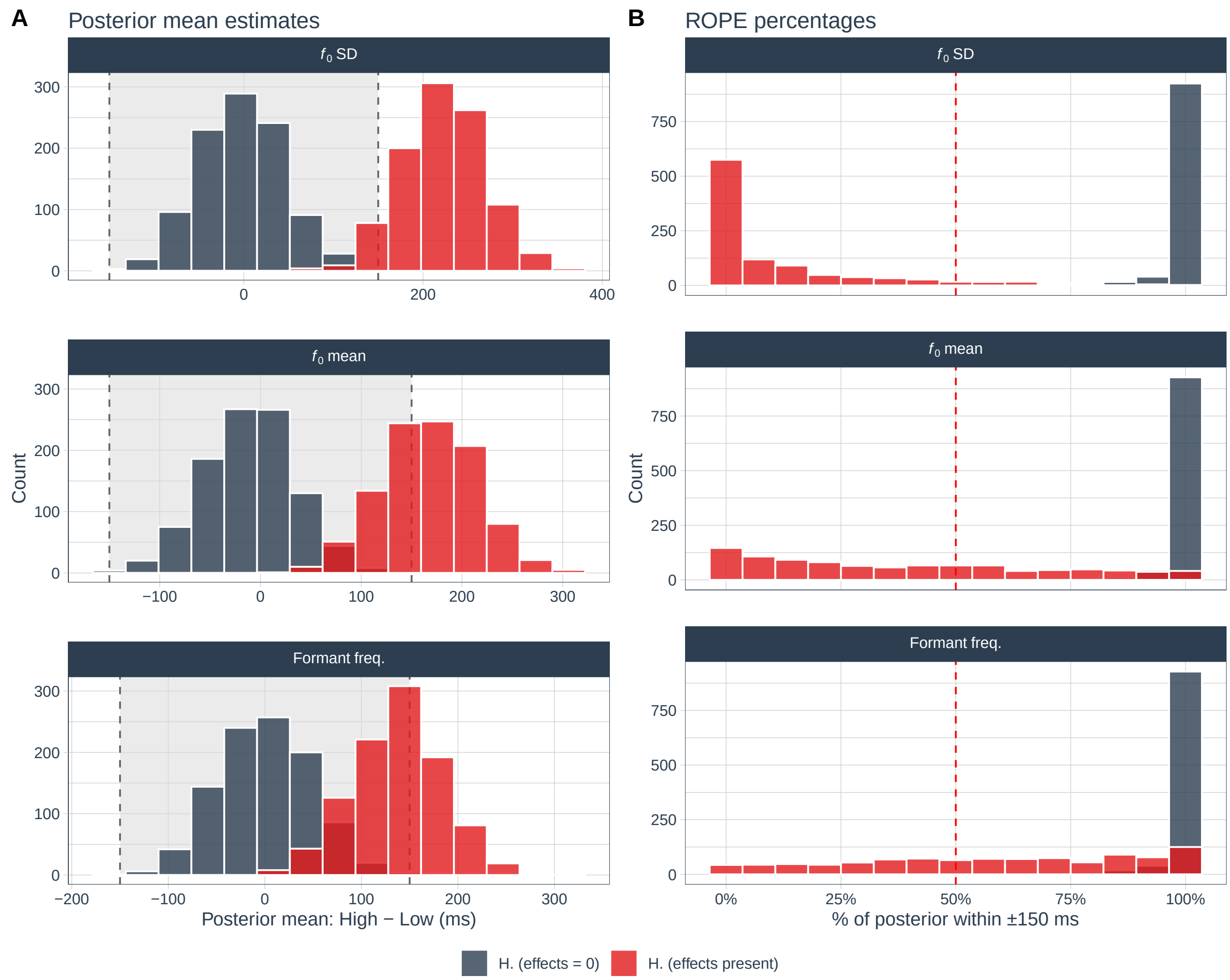

Figure S5. Bayesian posterior estimates and ROPE percentages across 1000 simulations at $n = 200$, under both populations. **Left column:** Posterior mean estimates (High − Low, ms). The grey shaded region marks the ROPE (±150 ms). Under H_1, posteriors should lie well outside the ROPE; under H_0, they should cluster near zero. **Right column:** ROPE percentages. The dashed line marks 50%. High ROPE % under H_0 and low ROPE % under H_1 confirms that the model discriminates correctly.

2.8 Overall Summary

```
detect_at_target |>
  left_join(equiv_at_target, by = c("term", "n_sample")) |>
  left_join(
    bayes_h1 |>
      group_by(term) |>
      summarise(`ROPE\\% (H$_1$)` = paste0(
                  round(mean(rope_pct, na.rm = TRUE) * 100, 1), "\\%"),
                .groups = "drop"),
    by = "term"
  ) |>
  left_join(
    bayes_h0 |>
      group_by(term) |>
      summarise(`ROPE\\% (H$_0$)` = paste0(
                  round(mean(rope_pct, na.rm = TRUE) * 100, 1), "\\%"),
                .groups = "drop"),
    by = "term"
  ) |>
  arrange(factor(term, levels = c("f0_sd", "f0_mean", "ffreq"))) |>
  mutate(
    Effect = case_when(
      term == "f0_sd"   ~ "$f_0$ SD (H1)",
      term == "f0_mean" ~ "$f_0$ mean (H2)",
      term == "ffreq"   ~ "Formant freq. (Exploratory)"
    ),
    `Exp. effect` = case_when(
      term == "f0_sd"   ~ paste0(eff_f0sd,    " ms"),
      term == "f0_mean" ~ paste0(eff_f0mean,  " ms"),
      term == "ffreq"   ~ paste0(eff_ffreq,   " ms")
    ),
    `Det. power`   = paste0(round(power_detect * 100, 1), "\\%"),
    `Equiv. power` = paste0(round(power_equiv  * 100, 1), "\\%")
  ) |>
  select(Effect, `Exp. effect`, `Det. power`, `Equiv. power`,
         `ROPE\\% (H$_1$)`, `ROPE\\% (H$_0$)`) |>
  tt(notes = paste0(
      "\\textit{Note.} Equivalence bounds: $\\pm ", eq_bound, "$ ms ",
      "(see \\hyperref[sec-bounds]{Section~2.3}). ",
      "Det. power: frequentist LMM, $p < .05$, H$_1$ population; ",
      "target $\\geq$95\\% for H1 and H2 (no constraint for exploratory ",
      "formant frequencies). ",
      "Equiv. power: 90\\% CI within equivalence bounds, H$_0$ population. ",
      "ROPE \\%: mean proportion of posterior within $\\pm ", eq_bound,
      "$ ms across ", n_sim_bayes, " \\texttt{brms} simulations; H$_1$ ",
      "values should be low (posteriors outside ROPE when real effects exist) ",
      "and H$_0$ values should be high (posteriors near zero under null). ",
```

```
    "See \\hyperref[sec-bayes]{Section~2.6} for full Bayesian results ",
    "including HDI diagnostics."),
  width = c(0.27, 0.12, 0.14, 0.14, 0.16, 0.17)) |>
group_tt(j = list("Frequentist" = 3:4, "Bayesian ROPE" = 5:6)) |>
style_tt(align = "lccccc") |>
format_tt(escape = FALSE)
```

Table S7. Combined frequentist and Bayesian power summary at the pre-specified target sample size ($n = 200$). Detection and equivalence power are from Monte Carlo simulations (H_1 and H_0 superpopulations respectively). ROPE percentages are means across 1000 Bayesian LMMs fitted via `brms` at $n = 200$. The study meets the $\geq 95\%$ detection power target for both confirmatory hypotheses (H1 and H2). Formant frequencies are exploratory and carry no power constraint. Equivalence power is modest but reflects the precision limits of the infant eye-tracking paradigm (see Section 2.3).

Effect	Exp. effect	Frequentist		Bayesian ROPE	
		Det. power	Equiv. power	ROPE% (H_1)	ROPE% (H_0)
f_0 SD (H1)	100 ms	99.7%	83.7%	11.2%	98.9%
f_0 mean (H2)	75 ms	96.2%	83.6%	38.4%	99%
Formant freq. (Exploratory)	50 ms	84.4%	82.8%	57.8%	99%

Note. Equivalence bounds: ±150 ms (see Section 2.3). Det. power: frequentist LMM, $p < .05$, H_1 population; target ≥95% for H1 and H2 (no constraint for exploratory formant frequencies). Equiv. power: 90% CI within equivalence bounds, H_0 population. ROPE %: mean proportion of posterior within ±150 ms across 1000 `brms` simulations; H_1 values should be low (posteriors outside ROPE when real effects exist) and H_0 values should be high (posteriors near zero under null). See Section 2.6 for full Bayesian results including HDI diagnostics.

The simulation confirms that:

- At n = **200**, detection power exceeds 95% for both confirmatory main effects (H1: f_0 SD; H2: f_0 mean) under the assumed population effect sizes. Formant frequencies are treated as exploratory and carry no detection power constraint.
- **Equivalence bounds** are set at ±150 ms, anchored to the precision limit of the design: the smallest effect detectable by a practically feasible infant eye-tracking study of this type, given $\sigma \approx 1000$ ms. This corresponds to 3% of the total fixation-time range and 150% of the largest expected main effect (Lakens et al., 2018; Simonsohn, 2015).
- **Equivalence power** at $n = 200$ is approximately 83% on average, meeting the 80% target.
- The **frequentist equivalence test** generalises TOST to LMM fixed effects via model-based `emmeans` CIs; detection and equivalence power are estimated from separate superpopulations as required for correct interpretation (Lakens, 2017).
- The **Bayesian ROPE analysis** uses `brms` with weakly informative priors and `bayestestR`'s `equivalence_test()` (Makowski et al., 2019). ROPE percentages are summarised alongside frequentist results in the table above; H_1 values confirm sensitivity (posteriors outside the ROPE when real effects are present) and H_0 values confirm specificity (posteriors inside the

ROPE under the null). Full posterior distributions and HDI diagnostics are in Section 2.6.
- **Interaction effects** are not powered at this sample size and are treated as exploratory throughout.

Session Information

```r
pander::pander(sessionInfo(), locale = FALSE)
```

R version 4.6.1 (2026-06-24)

Platform: x86_64-pc-linux-gnu

attached base packages: *stats*, *graphics*, *grDevices*, *utils*, *datasets*, *methods* and *base*

other attached packages: *cmdstanr(v.0.9.0)*, *posterior(v.1.7.0)*, *bayestestR(v.0.18.1)*, *brms(v.2.23.0)*, *Rcpp(v.1.1.1-1.1)*, *scales(v.1.4.0)*, *ggpubr(v.0.6.3)*, *PerformanceAnalytics(v.2.1.0)*, *quantmod(v.0.4.28)*, *TTR(v.0.24.4)*, *xts(v.0.14.2)*, *zoo(v.1.8-15)*, *tidyquant(v.1.0.12)*, *ggdist(v.3.3.3)*, *furrr(v.0.4.0)*, *future(v.1.70.0)*, *effectsize(v.1.0.2)*, *emmeans(v.2.0.3)*, *lmerTest(v.3.2-1)*, *lme4(v.2.0-1)*, *Matrix(v.1.7-5)*, *lubridate(v.1.9.5)*, *forcats(v.1.0.1)*, *stringr(v.1.6.0)*, *dplyr(v.1.2.1)*, *purrr(v.1.2.2)*, *readr(v.2.2.0)*, *tidyr(v.1.3.2)*, *tibble(v.3.3.1)*, *ggplot2(v.4.0.3)*, *tidyverse(v.2.0.0)* and *tinytable(v.0.16.0)*

loaded via a namespace (and not attached): *Rdpack(v.2.6.6)*, *rlang(v.1.2.0)*, *magrittr(v.2.0.5)*, *otel(v.0.2.0)*, *matrixStats(v.1.5.0)*, *compiler(v.4.6.1)*, *loo(v.2.9.0)*, *vctrs(v.0.7.3)*, *quadprog(v.1.5-8)*, *pkgconfig(v.2.0.3)*, *crayon(v.1.5.3)*, *fastmap(v.1.2.0)*, *backports(v.1.5.1)*, *labeling(v.0.4.3)*, *pander(v.0.6.6)*, *rmarkdown(v.2.31)*, *tzdb(v.0.5.0)*, *ps(v.1.9.3)*, *nloptr(v.2.2.1)*, *bit(v.4.6.0)*, *xfun(v.0.58)*, *jsonlite(v.2.0.0)*, *broom(v.1.0.13)*, *parallel(v.4.6.1)*, *R6(v.2.6.1)*, *stringi(v.1.8.7)*, *RColorBrewer(v.1.1-3)*, *parallelly(v.1.47.0)*, *car(v.3.1-5)*, *boot(v.1.3-32)*, *numDeriv(v.2016.8-1.1)*, *estimability(v.1.5.1)*, *knitr(v.1.51)*, *parameters(v.0.29.1)*, *bayesplot(v.1.15.0)*, *splines(v.4.6.1)*, *timechange(v.0.4.0)*, *tidyselect(v.1.2.1)*, *rstudioapi(v.0.18.0)*, *abind(v.1.4-8)*, *yaml(v.2.3.12)*, *codetools(v.0.2-20)*, *processx(v.3.9.0)*, *curl(v.7.1.0)*, *listenv(v.0.10.1)*, *lattice(v.0.22-9)*, *withr(v.3.0.2)*, *bridgesampling(v.1.2-1)*, *S7(v.0.2.2)*, *coda(v.0.19-4.1)*, *evaluate(v.1.0.5)*, *RcppParallel(v.5.1.11-2)*, *pillar(v.1.11.1)*, *tensorA(v.0.36.2.1)*, *carData(v.3.0-6)*, *checkmate(v.2.3.4)*, *reformulas(v.0.4.4)*, *insight(v.1.5.1)*, *distributional(v.0.7.0)*, *generics(v.0.1.4)*, *vroom(v.1.7.1)*, *hms(v.1.1.4)*, *rstantools(v.2.6.0)*, *minqa(v.1.2.8)*, *globals(v.0.19.1)*, *xtable(v.1.8-8)*, *glue(v.1.8.1)*, *tools(v.4.6.1)*, *ggsignif(v.0.6.4)*, *mvtnorm(v.1.4-0)*, *grid(v.4.6.1)*, *rbibutils(v.2.4.1)*, *RobStatTM(v.1.0.11)*, *datawizard(v.1.3.1)*, *nlme(v.3.1-169)*, *Formula(v.1.2-5)*, *cli(v.3.6.6)*, *Brobdingnag(v.1.2-9)*, *gtable(v.0.3.6)*, *rstatix(v.0.7.3)*, *digest(v.0.6.39)*, *farver(v.2.1.2)*, *htmltools(v.0.5.9)*, *lifecycle(v.1.0.5)*, *bit64(v.4.8.2)* and *MASS(v.7.3-65)*

References

Arel-Bundock, V. (2026). *Tinytable: Simple and configurable tables in 'html', 'latex', 'markdown', 'word', 'png', 'pdf', and 'typst' formats* [R package version 0.16.0]. https://doi.org/10.32614/CRAN.package.tinytable

Aung, T., Hill, A. K., Hlay, J. K., Hess, C., Hess, M., Johnson, J., Doll, L., Carlson, S. M., Magdinec, C., G-Santoyo, I., Walker, R. S., Bailey, D., Arnocky, S., Kamble, S., Vardy, T., Kyritsis, T., Atkinson, Q., Jones, B., Burns, J., … Puts, D. (2024). Effects of Voice Pitch on Social Perceptions Vary With Relational Mobility and Homicide Rate. *Psychological Science*, *35*(3), 250–262. https://doi.org/10.1177/09567976231222288

Ben-Shachar, M. S., Lüdecke, D., & Makowski, D. (2020). effectsize: Estimation of effect size indices and standardized parameters. *Journal of Open Source Software*, *5*(56), 2815. https://doi.org/10.21105/joss.02815

Boersma, P., & Weenink, D. (2026). *Praat: Doing phonetics by computer* (Version 6.4.63). http://www.praat.org/

Broesch, T. L., & Bryant, G. A. (2015). Prosody in Infant-Directed Speech Is Similar Across Western and Traditional Cultures. *Journal of Cognition and Development*, *16*(1), 31–43. https://doi.org/10.1080/15248372.2013.833923

Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, *80*(1), 1–28. https://doi.org/10.18637/jss.v080.i01

Bürkner, P.-C. (2018). Advanced Bayesian multilevel modeling with the R package brms. *The R Journal*, *10*(1), 395–411. https://doi.org/10.32614/RJ-2018-017

Bürkner, P.-C., Gabry, J., Kay, M., & Vehtari, A. (2026). Posterior: Tools for working with posterior distributions [R package version 1.7.0]. https://mc-stan.org/posterior/

Corretgé, R. (2012). Praat vocal toolkit [Praat plugin for voice manipulation]. https://www.praatvocaltoolkit.com

Cox, C., Bergmann, C., Fowler, E., Keren-Portnoy, T., Roepstorff, A., Bryant, G., & Fusaroli, R. (2023). A systematic review and Bayesian meta-analysis of the acoustic features of infant-directed speech. *Nature Human Behaviour*, *7*(1), 114–133. https://doi.org/10.1038/s41562-022-01452-1

Dancho, M., & Vaughan, D. (2026). *Tidyquant: Tidy quantitative financial analysis* [R package version 1.0.12]. https://doi.org/10.32614/CRAN.package.tidyquant

Field, A. (2013). *Discovering statistics using IBM SPSS statistics: And sex and drugs and rock 'n' roll* (4th edition). Sage.

Gabry, J., Češnovar, R., Johnson, A., & Bronder, S. (2025). *Cmdstanr: R interface to 'cmdstan'* [R package version 0.9.0]. https://mc-stan.org/cmdstanr/

Hilton, C. B., Moser, C., & Mehr, S. (2020, April). Human vocalization corpus: Recordings of infant-directed and adult-directed speech and song in 21 societies. https://doi.org/10.5281/zenodo.5525161

Hilton, C. B., Moser, C. J., Bertolo, M., Lee-Rubin, H., Amir, D., Bainbridge, C. M., Simson, J., Knox, D., Glowacki, L., Alemu, E., Galbarczyk, A., Jasienska, G., Ross, C. T., Neff, M. B., Martin, A., Cirelli, L. K., Trehub, S. E., Song, J., Kim, M., … Mehr, S. A. (2022). Acoustic regularities in infant-directed speech and song across cultures. *Nature Human Behaviour*, *6*(11), 1545–1556. https://doi.org/10.1038/s41562-022-01410-x

John K. Kruschke. (2015). *Doing Bayesian data analysis: A tutorial with R, JAGS, and Stan* (2nd ed.). Academic Press.

Kassambara, A. (2026). *Ggpubr: 'ggplot2' based publication ready plots* [R package version 0.6.3]. https://doi.org/10.32614/CRAN.package.ggpubr

Kay, M. (2024). ggdist: Visualizations of distributions and uncertainty in the grammar of graphics. *IEEE Transactions on Visualization and Computer Graphics*, *30*(1), 414–424. https://doi.org/10.1109/TVCG.2023.3327195

Kay, M. (2025). *ggdist: Visualizations of distributions and uncertainty* [R package version 3.3.3]. https://doi.org/10.5281/zenodo.3879620

Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. B. (2017). lmerTest package: Tests in linear mixed effects models. *Journal of Statistical Software*, *82*(13), 1–26. https://doi.org/10.18637/jss.v082.i13

Lakens, D. (2017). Equivalence Tests: A Practical Primer for *t* Tests, Correlations, and Meta-Analyses. *Social Psychological and Personality Science*, *8*(4), 355–362. https://doi.org/10.1177/1948550617697177

Lakens, D., Scheel, A. M., & Isager, P. M. (2018). Equivalence Testing for Psychological Research: A Tutorial. *Advances in Methods and Practices in Psychological Science*, *1*(2), 259–269. https://doi.org/10.1177/2515245918770963

Lenth, R. V., & Piaskowski, J. (2026). *Emmeans: Estimated marginal means, aka least-squares means* [R package version 2.0.3]. https://doi.org/10.32614/CRAN.package.emmeans

Makowski, D., Ben-Shachar, M. S., & Lüdecke, D. (2019). Bayestestr: Describing effects and their uncertainty, existence and significance within the bayesian framework. *Journal of Open Source Software*, *4*(40), 1541. https://doi.org/10.21105/joss.01541

Simonsohn, U. (2015). Small Telescopes: Detectability and the Evaluation of Replication Results. *Psychological Science*, *26*(5), 559–569. https://doi.org/10.1177/0956797614567341

Vaughan, D., Bengtsson, H., & Dancho, M. (2026). *Furrr: Apply mapping functions in parallel using futures* [R package version 0.4.0]. https://doi.org/10.32614/CRAN.package.furrr

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., … Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, *4*(43), 1686. https://doi.org/10.21105/joss.01686